Relativity and Electromagnetism Homework Assignment 2

Submission deadline: 2024 April 18 (please submit by 17:00 CEST)

[*Note:* This is the second homework assignment on the basics of electromagnetism. It contributes to 10% of the total points for final evaluation. You will be graded not only on the final result of a problem, but also on the reasoning and the intermediate steps. Please upload a (LaTeX / scanned) **pdf** copy of your answer sheet on Ufora by the deadline indicated above.]

Problem 1. A spaceship stavels to a star 8 light-years away, in a time its crew considers to be 8 years. What is the ship's speed? **[1 point]**

Problem 2. Consider two events whose spacetime coordinates (ct, x, y, z) in some inertial frame $\mathcal{S}$ are $(0, 0, 0, 0)$ and $(1, 2, 0, 0)$ respectively. Mark the two events on a conventional spacetime diagram. Find the speeds of the frames (with respect to $\mathcal{S}$) in which (i) the events are simultaneous, (ii) the second event occurs $ct = 1$ unit of time *before* the first event. Is there a frame in which the two events occur at the same point? **[2 points]**

Problem 3. *Relativistic Doppler effect.* In an inertial frame, a train of plane light waves of wavelength λ travels in a negative x direction towards the observer at the origin. The loci of the wavecrests then satisfy an equation of the form $x = -ct + n\lambda$ with n as intergers. Sketch some such loci on a spacetime diagram in a frame $\mathcal{S}$ in which the source is at rest. Apply a Lorentz transformation which takes you to a frame $\mathcal{S}'$ in which the source is moving to the left with a speed v. Deduce that

$$\lambda' = \sqrt{\frac{c-v}{c+v}}\lambda\,.$$

[2 points]

Problem 4. In a particle collider experiment, consider a proton and an antiproton each of rest mass 938 MeV/c^2 moving towards each other. Each of the two particles has a speed $\frac{\sqrt{3}}{2}c$ (in the frame of the collider). They collide and produce a bunch of pions of mass 134 MeV/c^2. [*Note:* This is a very unlikely outcome in a real experiment. But assume that this happens.] What is the maximum number pions which can be produced? **[1 point]**

Problem 5. Write down the transformation rules for the electric and magnetic fields (i.e. analogues of Eq. 12.109 of the textbook) for a Lorentz boost in the z direction. **[1 point]**

Problem 6. The electromagnetic field tensor is defined (in our conventions) as:

$$F^{\mu\nu} \equiv \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}.$$

a. Write down the components of the *covariant* field tensor $F_{\mu\nu}$ in a similar matrix form.

The dual field tensor $\tilde{F}^{\mu\nu}$ is defined as

$$\tilde{F}^{\mu\nu} \equiv \frac{1}{2}\,\varepsilon^{\mu\nu\lambda\sigma} F_{\lambda\sigma}\,,$$

with the convention $\varepsilon^{0123} = 1$.

b. It turns out that $\varepsilon^{1230} = -1$. Why?

c. Using the above definition explicitly calculate $\tilde{F}^{01}$, $\tilde{F}^{02}$, $\tilde{F}^{12}$, and $\tilde{F}^{13}$.

These results should be consistent with the matrix for $\tilde{F}^{\mu\nu}$ written down in class (or equivalently the matrix $G^{\mu\nu}$ in the textbook).

d. Write down an expression for $F^{\mu\nu}$ in terms of $\tilde{F}_{\mu\nu}$ and the $\varepsilon^{\mu\nu\lambda\sigma}$ tensor. *Note:* It is possible to work out this part of the problem avoiding nasty calculations.

[3 points]